

FPDAF

- **Review 1 (Problem Formulation & Concept Design):** We defined the research gap (privacy concerns, clinical demographic drifts, and black-box algorithms in ICU models) and proposed the conceptual mathematical framework of FPDAF—introducing bi-level model optimization and local CUSUM drift detection boundaries.
- **Review 2 (Data Engineering & Pipeline Audit):** We completed the data preprocessing pipeline on over 40,000 PhysioNet patient records, implementing sequential forward-fill imputation, leakage-free standardized scaling, and 24-hour sequence padding, all verified by an automated quality-assurance audit reporting script.

Here is the chronological step-by-step breakdown of what we did for **Review 3:**

- **Step 1: Implemented Baseline Models** Coded the standard baseline models (Centralized LSTM, FedAvg, FedProx, and Ditto Personalized FL) in PyTorch to establish our comparative performance benchmark.
- **Step 2: Coded the Proposed FPDAF Model** Developed our upgraded model architecture, combining a **Bidirectional LSTM backbone** (to extract multi-directional clinical sequences) with a custom **4-Head Temporal Self-Attention layer** (to calculate sequence-level saliency).
- **Step 3: Programmed the Drift Adaptation (CUSUM & CSSP)** Integrated the statistical Cumulative Sum (CUSUM) monitor to track validation loss residuals locally. Programmed the **CSSP mechanism** to automatically freeze the BiLSTM and attention layers when drift is detected, training only local classifier heads.
- **Step 4: Ran Training & Evaluation Loops** Executed the federated training cycles over the hospital clients, outputted model weight checkpoints (.pt), evaluated comparative metrics, and ran ablation tests to verify the impact of each module.
- **Step 5: Synchronized the SQLite Database** Programmed a database synchronization script (init_db.py) to load the trained PyTorch weights, run live inferences on clinical test sequences, and populate the SQLite tables with risk predictions, timeline charts, and attention explainers.
- **Step 6: Deployed the Full-Stack CDSS Workstation** Built the **FastAPI backend** to serve database queries and developed the **React Vite frontend** configured with Tailwind dark/light modes, Recharts vital lines, Sepsis warning dials, and animated federated telemetry.

Here is the clear explanation of the inputs and outputs of our application:



1. Our Input

The input is a rolling **24-hour sequence window of multivariate physiological data** for an ICU patient, consisting of:

- **Continuous Vital Signs:** Heart Rate (HR), Systolic Blood Pressure (SBP), Body Temperature, Oxygen Saturation (SpO₂), and Respiration Rate.
- **Sparse Laboratory Measurements:** Hourly blood work values (like White Blood Cell count, Platelets, Creatinine, and Lactate levels).
- **Static Demographics:** General patient parameters (Age, Gender, and Ward location).



2. Our Output / Prediction

The model processes this 24-hour clinical sequence and outputs three specific results:

- **Sepsis Risk Probability Score:** A real-time probability percentage (from 0% to 100%) indicating the likelihood of the patient developing Sepsis.
- **Bedside Alarm Alert (Stable vs. Critical):** A binary clinical notification flag triggered if the risk probability score crosses the safety threshold.
- **Explainable Temporal Attention Heatmap (XAI Output):** A timeline overlay showing the exact attention weights computed for each hour. This tells the clinician exactly **which specific hours/trends in the last 24h** (e.g., a drop in oxygen saturation at hour 15 and a spike in heart rate at hour 19) caused the high risk score.

Here is a simple, high-impact clinical definition you can use for your presentation:

- **What Sepsis is:** Sepsis is a life-threatening medical emergency caused by the body's extreme, systemic immune response to an infection, which ends up damaging its own tissues, blood vessels, and vital organs.
- **Why it is Dangerous:** If not diagnosed immediately, it rapidly progresses to **Septic Shock**, leading to multi-organ failure, a sudden drop in blood pressure, and high mortality rates.
- **The Clinical Time-Window Challenge:** Sepsis is notoriously difficult to detect early because its initial symptoms resemble general infections. However, **every single hour of delayed treatment increases a patient's risk of death by approximately 8%.**
- **Role of Our Project:** By continuously analyzing ICU vital signals, our **FPDAF** framework acts as an early-warning radar system, alerting doctors hours before a patient clinically deteriorates, giving clinicians a vital head-start to administer antibiotics and fluids.

